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TOPICAL REVIEW

The Role of VR in Education: A Two-Phase Review of Umbrella Reviews and Recent Primary Studies

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ABSTRACT The use of virtual reality (VR) in research and development has increased dramatically across various fields over the past decade, and education is no exception. To study the status and recent trends of VR research in learning, we present a novel two-phase systematic review: First, in a meta-umbrella review, we drew together the general essence of VR studies by summarizing 11 umbrella reviews in the field from 2017 to present. Second, we assessed more recent research trends by constructing a topic model of over 6530 primary studies published in 2023–2024. We found that VR has become increasingly popular, offering various benefits across different fields, including healthcare and education. In specific learning domains, VR has been used for chemistry and engineering laboratory training, gamified experiences, aviation training, and various school subjects such as history. VR in education was typically characterized by personalized immersive experiences that could adapt to the learner. However, the definitions of VR, especially concerning the amount of immersion, were varied. VR can motivate, control the environment, and help with difficult concepts. Long-term evidence of the effectiveness of VR training was still scarce. Learning standards or frameworks for the use and design of VR in education have not yet been used or formed and important aspects like immersion have not been fully utilized. The results indicate that, more thorough research with long-term data, standards, and clear operationalization of VR features are needed to truly establish the benefits and improvements VR can provide within the field of education and beyond.

INDEX TERMS Analysis, meta-umbrella review, topic model, umbrella review, virtual reality.

I. INTRODUCTION

Recently, there has been a rapid increase in the use of *virtual reality* (VR) and *augmented reality* (AR)—collectively referred to by the term *extended reality* (XR)—and of other immersive experiences in many disciplines, including healthcare, history, education, and psychology, to name just a few [1]. Virtual environments typically experienced through the use of technology, such as a desktop computer or a full-scale flight simulator [2], are commonly referred as VR (which is the terminology that has been generally applied in this article). VR plays an important role in research and

development (R&D) and the number of VR-related academic publications has more than doubled since the year 2000 [3]. The technical ability to generate extended reality experiences has existed for some time [4]. The key reason for their recent popularity has been the new development of different devices to better provide the experiences, such as more practical and affordable head-mounted-displays (HMDs), especially after the release of Meta’s Oculus Rift product and its numerous updated versions [5]. Such technologies offer affordable high-quality VR experiences and have been used in different fields, including education and training [6].

Indeed, VR and its use in education and learning is an interesting and expanding research field. Immersive technologies can provide access to inaccessible places and

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allow the learning of dangerous or complex situations in a safe and interactive way. VR can improve the learner's involvement and motivation and, for individuals with certain disabilities, can potentially provide new ways to interact and learn [6]. VR can be used for various educational purposes, from investigating concrete things such as human cells [7] to learning about abstract things such as teamwork and communication [8]. The pedagogical aspects of VR emphasize its possibilities to offer engagement and realism along with more interactive collaborative actions. VR can motivate and add enjoyment to learning, and it can offer personalized learning experiences that are tailored to each learner and their needs [9]. Learning in VR can also be tied to the concepts of *gamification* and *game-based learning* [9].

Thus, a thorough review was conducted to obtain a better view of the vast and rapidly evolving field of VR in education. Because of the expansiveness of the research field, there already exist numerous primary empirical articles as well as reviews and summaries of this work on a secondary level in the form of literature reviews and meta-analyses. These reviews themselves can be summarized at a tertiary level in the form of umbrella reviews and meta-reviews of meta-analyses. Our first purpose, therefore, was to analyze and summarize research findings on the meta-umbrella review level (for simplicity, we use the prefix “meta” even when there was just a conceptual summary and no statistical analysis of the independent umbrella review studies).

For the second part of this article, this depiction was updated by the identification of the most recent primary articles on VR in education and learning. The goal was to compare these articles with the previously summarized research to identify topical continuums and possible changes of research trends. For this purpose, a second set of primary articles was analyzed using topic modeling, which provided an explorative way to not limit the analysis of the research content with preliminary hypotheses and that could be used to easily categorize a large amount of textual data into emerging topics [10].

The goal of the research, therefore, was to summarize both past and recent research knowledge on VR in education. For this purpose, the following research questions were posed:

- RQ1 How has VR been conceptualized in recent umbrella reviews?
- RQ2 What were the main topics in recent primary studies?

II. CONCEPTUALIZATION OF VR

Virtual reality is a multifaceted concept, often depicted on a reality-virtuality (RV) continuum [11], which spans from the tangible real world to fully immersive virtual environments (Figure 1). This continuum is achieved through a display in less immersive and more immersive ways. Between tangible reality and a virtual one lies *mixed reality*, where one can still see aspects of the real world [11]. The loosely defined “mixed reality” is used as a general term for different forms of

reality-enhancing methods. These include *augmented reality* (AR), where computer graphics are overlaid with the real world, and *augmented virtuality* (AV) in which most of the content is virtual, but there still exists some aspects of the real world. Finally, in virtual reality (VR), the real environment is fully replaced by a virtual one [5], [11]. Recently, extended reality (XR) has emerged as a broader term, in particular to refer to both AR and VR applications [12].

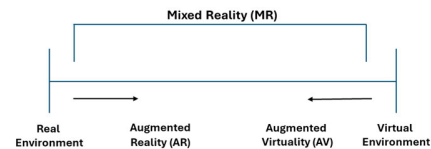


FIGURE 1. Milgram and Kishino's reality-virtuality (RV) (adapted from [11]).

The reality-virtuality (RV) continuum has recently been re-examined, and limitations have been noted, especially regarding its focus on visual aspects of realism over other senses [13]. For example, VR could also be considered “mixed reality” because of sensory conflicts, and “true” virtuality might only be achievable by directly connecting a brain to a simulation. Thus, newly proposed conceptual models include more than two dimensions and combine aspects such as real-world experience, immersion, and coherence to describe virtuality.

On the reality-virtuality (RV) continuum, this review was mostly concerned with virtual reality (VR) and how it has been used and defined in recent research. VR can be further divided into immersive virtual reality (iVR), in which the user is engaged with the virtual through a head-mounted display, and non-immersive virtual reality (n-iVR), in which the environment is depicted on a screen as in a video game [14]. Interaction in iVR is more real-time and is usually more connected to the user's movement through sensors or other means of control. In contrast to this, n-iVR is simpler and less connected to the user through simpler control methods such as a mouse or game controllers. iVR surrounds the user and thus immerses them in the environment [15].

One of the key aspects of virtual reality is its *sense of presence* and *immersion* [16]. Immersion could be said to be the element that draws in people and describes their engagement with the content during the experience. Video games are often immersive and take total control of a player's attention. Immersion can be described as having different levels, from mere engagement to total immersion. With total immersion, the real world seems to disappear and time passes quickly; the sensation is often compared to a flow experience [17].

Many things, such as the visual presentation and environment of a VR experience, can affect a feeling of immersion and presence both positively and negatively [14]. VR can, for example, affect cognitive load with its many different interactive elements [18]. Audio and visual elements can distract a user, and their cognitive skill acquisition can

be affected by the level of immersion. Immersion can even become a distraction to the learning process when certain elements of the environment are too interesting. Another negative effect related to immersion is the display of symptoms similar to those found with motion sickness, which may be due to unfamiliarity with headsets [14].

VR can be used to visualize or simulate complex situations and new ways of interaction [6]. For instance, virtual agents in VR can coach soft skills [19], and participants can be exposed to their fears in safe VR environments [20]. A robotic arm can be combined with VR to create a digital twin of a human model [21], and pedestrians' understanding of the designed features of autonomous cars can be examined within VR [22]. In the field of learning, there have been experiments related to how the gender of a teacher avatar can affect learning [23] and how distance learning can be enhanced with virtual environments [9].

Despite better overall accessibility, there are still many obstacles to the use of VR. There is a great deal of overhead associated with the procurement and use of VR equipment. There are software needs and costs. In many cases, modifications are required for the classroom use of VR, and it requires the continuous training of technical and maintenance personnel. VR adoption rates are still relatively low, so companies may not be ready to dedicate their resources to VR projects. Specialized hardware requirements and potential usability and input problems can be taxing and demanding, especially in the context of teaching and education [9], [14].

III. MATERIALS AND METHODS

Because of the many different review types that are touched upon in this article, a brief introduction on some of them is in order [24]. The goal of a systematic review is to pinpoint what is already known and what remains unknown and should be studied further. A systematic literature review is based on a comprehensive search strategy and the use of explicit inclusion and exclusion criteria. An umbrella review summarizes a set of systematic reviews [25]. After noting the number of available umbrella reviews, we decided to analyze them first, in other words, to realize a meta-umbrella review, thus establishing a big picture of the research field [26].

The umbrella reviews whose contents were analyzed were published between January 2017 and March 2024. We specifically chose to exclude earlier possible reviews as we did not want to include literature on more primitive versions of various VR technologies. The second phase of the literature review process realized **topic modeling of recent primary studies** using bidirectional encoder representations from transformers topic (BERTopic), a machine-learning model for natural language processing that supports topic modeling clustering document to individual topics accounting for the context and sentences in a broad way [27]. The search included primary studies published in 2023 and March of

2024, with a goal of providing a better view of the most recent virtual reality research on learning and education.

Both review phases were conducted according to the preferred reporting items for systematic reviews and meta-analyses (PRISMA) framework [28]. The overall two-phase review process is illustrated in Figure 2. The first phase covers RQ1 and the analysis of umbrella reviews aggregated into a meta-umbrella review. The second phase answers RQ2 with an analysis of recent primary-level studies through topic model analysis, abstracted into a systematic review.

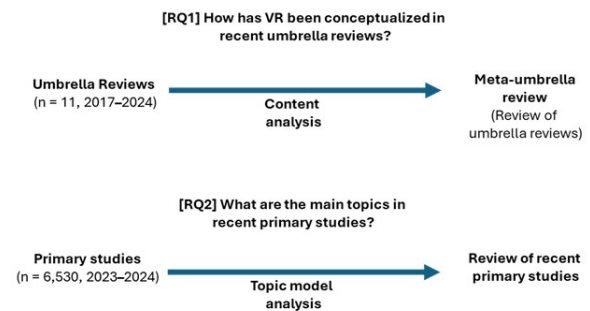


FIGURE 2. Description of the research process and results.

A. SEARCH STRATEGY AND SELECTION CRITERIA

Virtual reality is an active research topic, and a preliminary search for review articles from 2017 to 2023 for the keywords “virtual reality” and “review” yielded over a thousand results (Scopus $n = 1,228$; Google Scholar $n = 2,430$). Thus, the first phase of our research focused on umbrella reviews (Figure 2). The initial search for umbrella reviews was conducted in November 2023 and focused on titles, abstracts, and keywords. A complementary search was conducted in March 2024, and the final set of 11 umbrella reviews contained articles from 2017 through March 2024. The search string for obtaining umbrella reviews was

TITLE-ABS-KEY (“virtual reality” AND (“tertiary” OR “meta review” OR “umbrella review” OR “meta analysis”))

The search string was adjusted according to different databases and their standards (e.g., the publishing year was added to the phrase or determined in the search tool). The databases searched were Scopus, Web of Science, Pubmed, ProQuest, SAGE Journals Online, ScienceDirect (Elsevier), Taylor & Francis, Wiley, ACM Digital Library, IEEEExplorer, and SpringerLink.

For the second phase, the search was targeted at recent empirical studies published from 2023 until March 2024 (Figure 2). The topic modeling of these studies aimed to provide a broad overview of recent trends in the research, complementing the analysis of the umbrella reviews. This search was conducted using Scopus because it noted as having wide coverage of the academic literature in terms of publication channels and subject areas [29]. “Learning” and “training” were added as keywords to cover a wide

range of different settings. The search string for obtaining primary studies—in other words, conference papers (“cp”) and articles (“ar”)—was as follows:

(TITLE-ABS-KEY (“virtual reality”) AND (TITLE-ABS-KEY (education) OR TITLE-ABS-KEY (learning) OR TITLE-ABS-KEY (training))) AND PUBYEAR > 2022 AND PUBYEAR < 2025 AND (LIMIT-TO (DOCTYPE, “cp”) OR LIMIT-TO (DOCTYPE, “ar”))

B. SELECTION PROCESS

After searching the previously listed databases, the umbrella review search yielded 1,724 studies: Scopus (n = 701), Web of Science (n = 478), Pubmed (n = 26), ProQuest Computer database (n = 212), ScienceDirect (Elsevier) (n = 25), Taylor & Francis (n = 19), Wiley (n = 107), ACM Digital Library (n = 18), IEEE Explorer (n = 35), and SpringerLink (n = 103). Duplicate documents were identified and removed (n = 490). The remaining unique articles were screened by the primary author using the title, abstract, and keyword of each. This process is depicted in Figure 3. An article was included when it was an umbrella review or a review of reviews related to VR, accessible, and in English. An article was included in the meta-umbrella review when it 1) was an umbrella review or a review of reviews, 2) the topic of the article related to VR, 3) the full text was available, and 4) the article was written in English. For a more general view, the search was not restricted to any field or area. After the screening, 1,223 studies were excluded, and the relevant studies (n = 11) were kept for the full-review phase.

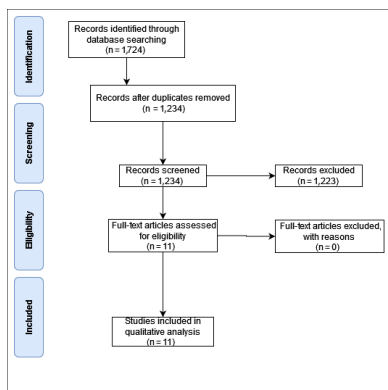


FIGURE 3. PRISMA flow chart for the meta-umbrella review.

The primary studies for topic modeling covered articles related to learning and training in VR in different fields (n = 6,530). Some topics may not have been VR-related prior to the detailed screening, but this was addressed in the analysis phase. Surprisingly, even though reviews had a dedicated document type (“re”) in Scopus, there were approximately 250 reviews included in the over 6,000 primary studies. Because of their relatively small proportion, they were kept in the set of articles that were topic modeled.

C. DATA ANALYSIS

For the **meta-umbrella review**, factors concerning metadata and VR conceptualization were first defined. The metadata of interest were defined as *authors, title, journal, topic, keywords, publishing years of reviews, VR technologies, and research methods*. Some metadata such as authors and journals were collected more for the maintenance of the database and for the possibility of analyzing how connected the different authors and journals could be. In the end, these were only a small part of the analysis process. The main analysis framework for the conceptualization consisted of *virtual reality definition, VR benefits and challenges, immersion, and pedagogical principles*. Two of the authors first collected this information into a table, and this content was discussed, elaborated, and analyzed by the entire group of authors. The results of this phase are presented in Section IV.

The analysis of the articles for **topic modeling** was mostly carried out by the primary author with feedback from the other authors. The total number of the analyzed documents was 6,282. The collected metadata were *authors, titles, year, source title and publishing specifics (journal, volume, DOI etc.), abstracts, author keywords and document types*. This data was compiled into an CSV file for easy extraction of data. The abstracts, keywords, and titles were extracted into an array where one index element had the abstract, keywords and title of each document. From this data BERTopic was used to form the topic model. BERTopic uses transformers and embeddings to generate new insights from a large set of data [30]. For the transformer we used the default model “all-MiniLM-L6-v2”, which works especially for English documents [27]. The documents were clustered with Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) algorithm and the class-based Term Frequency-Inverse Document Frequency (c-TF-IDF) was used to generate the topics [30].

For the topic model itself, a minimum topic size was set to 30 documents, resulting in 38 topics with 2,175 outliers identified by BERTopic (they were assigned a topic category of −1), including English stop words and various other smaller words. These were reduced by BERTopic’s distribution strategy “CountVectorizer”, which preprocessed the documents after the embed and cluster generation, thus spreading these representations to other topics [27]. This added some stop words to various topics, which were then removed while the length of the topic representations were adjusted with the *reduce outlier* method and the “*distributions*” strategy. When depicting the contents encapsulated by BERTopic, we used the most representative documents of different topics chosen by BERTopic.

IV. GENERAL VIEW OF VR ACCORDING TO RECENT UMBRELLA REVIEWS

This section presents the results of the meta-umbrella review. First, a small overview is provided, and after this,

the collection, analysis, and comparison of the results are described. The full analysis table is given in Supplementary Material, and a summary overview is depicted in Figure 4.

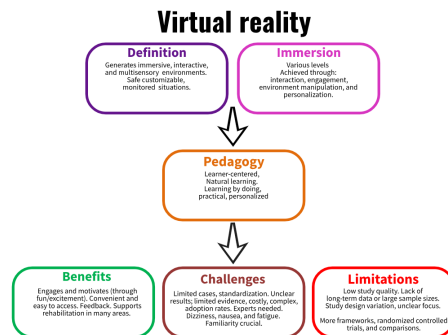


FIGURE 4. VR pedagogy based on the umbrella reviews; the full analysis and topics are depicted in the Supplementary Material.

A. OVERVIEW OF THE INCLUDED STUDIES

All of the umbrella reviews were published in 2020 or after: two in 2020 [31], [32], one in 2021 [33], one in 2022 [34], four in 2023 [35], [36], [37], [38], and two in 2024 [39], [40]. The various individual reviews that were the focus of the umbrella reviews were from 2000 and later; many of these individual reviews used 2000 as the beginning point of included literature as in [41]. In some of the umbrella reviews, a search was made from a single database without explicitly reporting of the years covered, as in [33] and [35]. The journals in which the reviews were published were quite well known and overall seemed to have good impact factors, indicating the quality of the reviews. These journals were: *The Electronic Library*, *Journal of Neuroengineering and Rehabilitation*, *Journal of Clinical Medicine*, *PeerJ*, *Frontiers in Medicine*, *Journal of Medical Internet Research*, *Age and Aging* (two reviews), *Current Problems in Cardiology*, *JMIR Serious Games*, and *Education Sciences*.

A summary overview of the umbrella reviews and analyzed areas is depicted in Figure 4. As the names of the journals readily indicate, the umbrella reviews mainly focused on healthcare and rehabilitation or therapy. The two exceptions most relevant here dealt with the effectiveness of teaching in extended reality (including VR, MR, and AR) [41] and distance learning in higher education using augmented and virtual reality (only VR) [32]. Both of these reviews contained aspects on student performance, though in [32], the focus was more on the design of remote learning to improve student engagement. The main interests in [41] were student achievement, learning motivation, academic achievement, and learning skills.

Most of the umbrella reviews were rehabilitation related. Four of these focused on physical rehabilitation: One review by Yu et al. [35] focused on the efficacy of training and rehabilitation in VR for Parkinson's disease by using relevant variables such as motor performance. Others, such

as Voinescu et al. [33], studied how VR-based interventions could help with the rehabilitation of impairments, collecting data on symptoms, interventions, VR technology, and study design. Eost-Tealling et al. [39] investigated the prevention of falls for dementia patients, collecting data related to technology, rehabilitation, patients, and cognition. A review by Jia et al. [37] focused on cardiac rehabilitation, collecting expansive data on technologies and physical and mental health.

The second main set of five umbrella reviews discussed on mental health or therapies in different contexts. Harith et al. [34] focused on digital mental health interventions, their effectiveness, and data collection. Viderman et al. [36] tackled VR for pain management, systematically focusing on anxiety and VR. Hu et al. [38] had a wide scope on nursing practice and patient care with pediatric, surgical, and gerontological care, focusing especially on the uses and benefits of VR. Yu et al. [40] focused on specific cognitive and mental aspects of elderly care, collecting data of inspected groups, VR, cognitive functions, attention, memory, executive functions, etc. Lastly, Dellarizzo et al. [31] collected evidence on VR-based therapies for psychiatric disorders, especially anxiety, with a broad search for meta-analyses investigating VR efficacy.

All the umbrella reviews evaluated limitations in the inspected VR research and concluded that the quality of VR research varied a great deal. For example, in one of the reviews, only three of the inspected articles were of moderate quality, and the rest were rated lower [35], with gaps in evidence, low methodological quality, and a lack of long-term data. Voinescu et al., Eost-Telling et al., Yu et al. [33], [39], [40] also noted a lower-than-moderate quality of articles. The research design in the studies was also limited, especially concerning the use of pre-tests, post-tests, and control groups; the research did not seem to take into account some important aspects of VR such as immersion [33], [39], [40]. The lack of sufficient information concerning research focus, methodological data, integration of VR, and how to ideally use VR were mentioned [31], [41]. One review also noted the limited number of comparisons of different technologies and their effectiveness [37]. Overall, a great deal of VR research has been conducted, but its quality and in particular deeper effects have not been properly examined.

B. DEFINITIONS AND IMMERSION

The definition or concept of VR varied a great deal among the reviews. At times, a formal definition was missing or only mentioned briefly through VR-related aspects such as immersion. Most reviews mentioned immersive virtual reality (iVR), which focuses of immersion through a head-mounted display and its aspects, but again the definition and focus varied. Usually VR was defined as a way to generate virtual images to achieve some type of immersive effect [31], [32], [36], [41], though this level could easily vary from iVR solutions as in [41] to mostly focusing on depicting the environment on a screen with non-immersive virtual reality

(n-iVR) solutions with simple desktop or motion-based games only briefly mentioning VR [39]. Immersion and a sense of presence was also often mentioned [31], [37], [38], [40]. VR was also often seen as a great way to safely experience and manipulate the environment and simulate real-world objects, events, and interaction [31], [32], [37], [38]. VR could also be used to provide real-time feedback, and it could be used to more easily standardize treatments or other solutions and personalize the experience for each user's needs and performance [38], [40].

The wide reality-virtuality continuum of VR was raised in many reviews. Most reviews included both iVR and n-iVR and often did not touch upon how these could differ from each other [35], [39]. One review even explicitly mentioned how VR technology is usually understood as a more immersive HMD experience, but the authors had decided to use a broader definition that included laboratories, mobile devices, and PCs [32]. One interesting outlier was a review focused on digital mental health [34], in which Internet environments were seen as a (kind of) virtual environment. This certainly would not be called immersive, but it highlights how the concept of virtuality is seen and how this can make it challenging to find suitable examples or results, especially when looking for more immersive and modern solutions.

C. RANGE OF VR SOLUTIONS

The Conceptualization and definition of VR is directly linked to views of technology. Indeed, perspectives on VR technology varied a great deal among the different umbrella reviews. Most of the articles mentioned VR headsets and their benefits or use, but some, such as [32] and [40], included less immersive solutions in the reviews or included no definition of VR [39]. Most of the other reviews also listed simpler exergames, such as Nintendo Wii, as simply VR even though these should be considered n-iVR [33], [38], [40]. However, some reviews were more iVR focused, such as [38].

The umbrella reviews used various definitions for immersion. Often, it was not clearly defined or even mentioned, so it had to be gleaned from the technologies or descriptions that were used. Immersion is connected to the technology used; the better this technology, the better the results that can be achieved to create authentic worlds [36]. Often the technology in the reviews was focused on less immersive solutions, such as Nintendo Wii and Microsoft Kinect [35], [37], [39]. Immersion can enhance treatments and help obtain better results in areas such as pain treatment, and immersion can be easily affected by factors such as tactile feedback and motion tracking [36], [37]. Immersion can also be affected by the customization and personalization of the environments [31], [33]. A full-flow immersive rehabilitation in which everything is based on the patient's needs in particular could be quite effective [40]. In the same vein, interaction of the environment can affect immersion and engage users more thus promoting treatments and learning [32], [33]. Fun and flexibility was also seen as an important part of immersion [35].

D. BROADER INTERPRETATIONS ON CONCEPTUALIZATION OF VR

As demonstrated by these articles, it seems that virtual reality can be seen as many things, but often, formal definitions or descriptions of VR and its aspects were missing or lacking. Usually, VR was represented as a way to create immersive worlds, but this was extended to mean that even simpler solutions were VR. Some reviews emphasized how important immersion, interactivity, feedback, and personalization could be. iVR was often used, but most of the technology was a simpler solution in VR or desktop VR. These can be called virtual environments, but they are easily confused with real iVR, making the benefits of both difficult to distinguish. Mostly, the reviews contained a mix of the various levels of virtuality, and there was not one that focused strictly on iVR.

E. BENEFITS AND CHALLENGES OF VIRTUAL REALITY

All the umbrella reviews listed many benefits and challenges of VR, from varied examples of entertaining learning to potential technical problems. Overall, the benefits seemed to outweigh the negatives, though often this effect was not that large. For example, learning with VR can motivate students and improve academic performance with small to moderate effects [41]. VR can be entertaining and motivating, and in particular, iVR can be more exciting than ordinary interventions [35]. For rehabilitation, VR seemed to have positive effects for those with Parkinson's disease [35] and for pain treatment [36]. VR also helped with fall prevention and had benefits with regard to cognitive functions and health [39]. VR and other digital interventions are accessible and easier to schedule [34], [37]. Even just adding VR can engage and have a measured impact on performance especially for those with phobias and neurocognitive disorders [31].

The VR studies did lack more concrete evidence, the transfer of VR skills to real life, and how technical limitations affect it are a mystery [33]. Research with VR is susceptible to a novelty effect, which can particularly affect small studies [32], [35], [41]. VR may also cause some adverse effects, such as dizziness or even vomiting [33], [38], though these can also be the results of traditional methods [40]. Due to the technology involved, VR can be a costly endeavor [37], [41]. This contributes to interventions with durations that are too short to obtain meaningful results and experimental groups that are too small [39]. There was a great deal of variation in the VR results for general aspects such as stress; for example, despite mixed results, VR fared better with individuals in the treatment of phobias [31], [34], even though the results were low compared to other methods.

Research on VR also seems to lack guidance, and standards are missing or still being developed; as previously mentioned, the evidence that has been presented in this area can be quite inconsistent [37], [38], [41], and there are some concerns about methodological quality about the research in this area [37], [40]. VR in itself is not a panacea, and traditional methods are still needed. How this

new technology is used is most important, and customized interventions could help provide better results [33]. Levels of technology acceptance can have a big impact on the adoption of new interventions, such as VR, especially with older individuals [39]. VR may not fit every situation; to implement VR and use it effectively, teachers need guidance, frameworks, experience, and interest [32], even when the technology would seem to be able to easily add entertainment and engagement in a classroom [41].

Overall, there seems to be a lack of more concrete and long-term evidence regarding VR interventions. When compared to traditional interventions, VR has been shown to produce more positive or at least comparable results. There are still many gaps in the literature, regarding the novelty of the VR, the effect of guidance, and more complete environments, among others. VR does seem to be able to provide a number of benefits in the way it makes interventions and rehabilitation more real-time, personal, and reactive for a patient, but the knowledge base associated with traditional methods is still needed.

F. ASPECTS OF LEARNING

In most of the analyzed umbrella reviews, the learning or educational aspects, such as how learning was seen or how people learn with VR, were difficult to assess directly. Most reviews were not directly connected to learning or education. Many of the reviews were more focused on intervention or rehabilitation; however, they often had learning or educational aspects in their methods or training of the patient. For this purpose, the concept of learning and education was examined in broad terms. Rehabilitation and training can involve various forms of learning, especially for an individual to contribute to their recovery process and required adaptations [42]. This dynamic process can integrate physical, cognitive, emotional, and behavioral aspects in promoting a patient's quality of life.

In a rehabilitation context, immersion and interaction interacted with positive educational aspects of VR, such as the ease of repetition and personalization with VR training methods for each individual patient [33]. Rehabilitation training with VR was seen as an immersive, convenient method with feedback possibilities [35]. This focus on feedback, immersion, and personalization also came up in other reviews [37], [38], [40]. Some pedagogical aspects could also be seen in how pain management in VR focused on the interaction and use of the multi-sensory immersive environments [36]. VR therapies with interaction and environment manipulation were seen as more learner based, able to adjust to the learners' needs, and there were even descriptions on transfer and how an individual could learn more with VR [31].

Two reviews on VR education described the nature of the interactive learning. One focused on teaching effectiveness, with natural and interactive learning with the help of iVR environments [41]. The other review focused on remote

learning in VR, and how this could impact performance and engagement with a focus on VR as an interactive simulation [32]. There was also some focus on user-centered design when the benefits of the active involvement of the user in design, personalization, and implementation of the environment [39]. Even reviews that were less VR focused had learning aspects in how retention and effectiveness of learning was affected by user engagement and experience [34].

Including these intervention and rehabilitation methods in our education view added multiple pedagogical aspects of VR. The umbrella reviews represented VR learning as practical and engaging. It provided abstractions of complex concepts and easy-to-understand visualizations. These concepts of interactive learning, learning through experience, and learning by doing were raised multiple times. Most of the umbrella reviews mentioned immersion and engagement and their effect on learning or rehabilitation methods. Many also noted the concepts of interaction, feedback, and personalization of experiences. There also seemed to be an overall focus on immersive learning in which personalization in design and interaction was important.

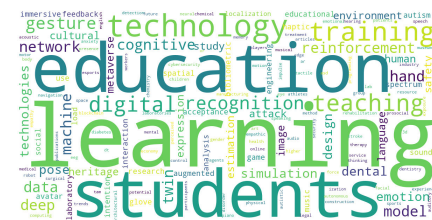


FIGURE 5. Word cloud for all topics.

V. MAIN TOPICS ACCORDING TO RECENT PRIMARY STUDIES

In this section, the results of the topic modeling concerning recent primary studies are presented. The topic words and representative documents chosen by BERTopic are inspected and analyzed. We encountered 38 different topics from 6,530 articles. The combined word cloud of all the topics is depicted in Figure 5. In additional figures other main topics are illustrated with word clouds, which were formed from the top ten words of the respective topics. Because of the search string used to locate the articles, all the word clouds originally contained “VR,” “virtual,” or “reality” as the most occurring words, but these were removed from the clouds for clarity and to make the differences among them more distinctive. “Learning”, “education,” and “students” were the most common words in these topics with a focus on different technologies and training methods from the metaverse to emotions. Different environments also had a clear impact, with “laboratories”, “simulations,” and “training” appearing. The topics also had a focus on many different subjects, areas, and study design.

Topics and their hierarchical division into thematic clusters (or subtrees, see Figure 6) were analyzed next in more detail

from an educational perspective. The topic tree had two main branches and four thematic clusters in total. Two of these were more outliers with a focus on technologies and the metaverse, so these were inspected only briefly. The more education- and learning-related clusters displayed in cyan and dark red in Figure 6 had many different aspects. One subset in red focused more on individual learning or experiences, while a subset in cyan had a focus on learning in groups.

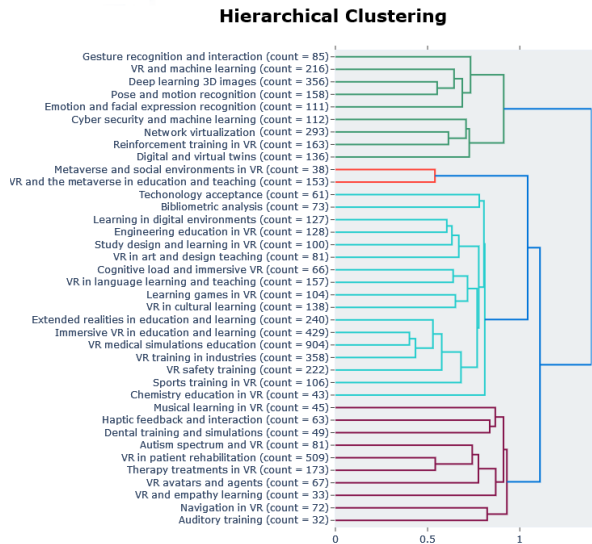


FIGURE 6. Hierarchy of all topics with their counts.

A. VR-RELATED TECHNOLOGIES

Nine topics in the first, green part of the tree in Figure 6 refer to the collection of experimental data from VR environments its analysis, and data-driven modeling, as well as data transfer and storage. This topic also had few outlier topics, such as *deep learning 3D images* and *reinforcement training in VR* [43, see], which focused on machine, not human, learning. *Network virtualization* was also related to the training and virtualization of data and networks, not humans. Similarly, the rest of the topics in this subset were more related to the multi-modal data and its analysis techniques when studying learning in VR, such as *VR and machine learning*, *pose and motion recognition*, *emotional and facial expression recognition*, and *digital and virtual twins*.

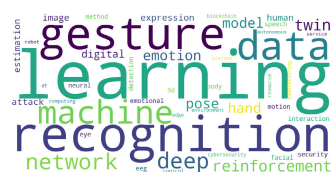


FIGURE 7. “VR-related technology” word cloud.

B. VR AND THE METAVERSE

The smallest and most narrow subset of topics in red in Figure 6 had only two topics related to the metaverse.



FIGURE 8. “VR and the Metaverse” word cloud.

The representative document of *VR and the metaverse in education and teaching* explored the possibilities of a new “edu-metaverse” ecosystem network model for innovative, accessible learning [44]. The focus was on how to use these immersive modern teaching environments with experimentation and continuous, engaging learning. The other topic *metaverse and social environments in VR* focused more on the legal aspects of the metaverse more than on learning [45]. It was interesting that the metaverse came up as its own tree, which indicates its growing importance in the field of VR.

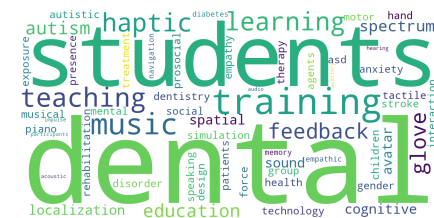


FIGURE 9. “VR for individual learning” word cloud.

C. VR FOR INDIVIDUAL LEARNING

The second largest sub-tree (dark red in in Figure 6) encapsulated 10 varying topics related to VR technologies (*haptic feedback and interaction, auditory training, VR avatars and agents, navigation in VR, and autistic spectrum and VR*), particular disciplines or use cases (*music learning in VR, dental training and simulations, VR in patient rehabilitation, and therapy treatments in VR*), or concerning an affective perspective (*VR and empathy learning*). Most of the topics were at least rehabilitation or intervention related so many of these were investigated in more detail. The topics related to technologies were mostly not analyzed; for example, the topic and the representative document on auditory learning did not seem to focus on VR [46].

Musical learning in VR had words such as “*music*,” “*musical*,” “*VR*,” “*teaching*,” and “*learning*”. The representative document was focused on how to build music animations and how the form of music elicits certain emotions: A music-based emotion recognition model and a musical animation system were constructed, which helped the user to understand music on a deeper level by generating unique personal music portraits [47].

The representative article on *dental training and simulations* was about recent VR developments and haptic implementation in dental education, focusing on schools' satisfaction and usage rates [48]. Most schools were satisfied

with the VR device implementations described, though some had lower integration rates. The usage of VR was high, but there were challenges with curricular changes and teacher acceptance.

Rehabilitation also came up in the topic of *autism spectrum and VR*, focusing on advancing the social abilities of ASD (autism spectrum disorder) children with VR therapy [49] and in *VR in patient rehabilitation*, on cognitive and motor skills. The representative article of the first topic was about assessing home-based VR and upper limb function rehabilitation for children with brain injuries through motion exercises and performance [50]. The home exercises proved motivational increasing movement, but the technology was lacking, the sample size was small, and there were no significant differences between the groups. For the second topic in [49], tailored experiences, adaptation, and real-time biofeedback were emphasized for engagement and reducing anxiety. The use of VR also benefited from intuitive interfaces and simplified controls, with proper integration and guidance from professionals.

The therapy orientation continued in *therapy treatments in VR* with mental health and anxiety, with the words such as "anxiety," "vr," "therapy," "treatment," and "mental". The representative article compared three intervention groups and investigated the role of VR in a personalized, relaxing iVR scenario and how it could impact anxiety and stress [51]. All the groups had a number of sessions with introductions to different areas related to the concepts, training, and a follow-up. The initial results seemed promising but had not been finalized.

VR in empathy learning emphasized design and social aspects of empathy with words such as "*empathy*," "*students*," "*prosocial*," "*VR*," and "*design*". The representative document focused on training pre-service teachers' empathy skills with the help of 360-video VR, and compared this to other solutions such as 2D video [52]. The VR resulted in higher empathy levels, but the results for perspective-taking were not significant. It seemed that the VR learning needed to be combined with instructional and scaffolding design for better learning results.

The topics in this tree indicate the many ways in which VR can be used in different contexts related to rehabilitation or education methods. Once again, VR seemed to have a positive impact with its interaction, immersion with personalization, and interactive environment. However, the results were varied, not all the factors were positive, and the sample sizes were lacking. VR may still not be well known in some contexts, such as schools, or it may need integration with a clear and comprehensive design to achieve the best results.

D. VR FOR LEARNING IN GROUPS IN SPECIFIC FIELDS

The largest cluster of topics colored cyan in Figure 6 recapitulated almost the entire field of our research. There were an outlier of specific studies (*Bibliometric analysis*), an individual topic related to *learning games in VR*, a small

group of specific topics related to the deployment and use of VR in general (*technology acceptance and cognitive load*), and many topics in the context of specific disciplines (*VR in art and design teaching, VR in language learning and teaching, VR medical simulations education, chemistry education in VR, sports training in VR, VR safety training, and engineering education in VR*). Moreover, the general topics *extended realities in education and learning and learning in digital environments* emerged with a focus on higher levels of abstraction about learning and studies on it in the extended reality of immersive VR and other environments.

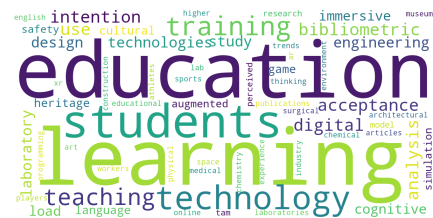


FIGURE 10. “VR for learning in groups in specific fields” word cloud.

The topic of *technology acceptance* focused on VR acceptance, usage, and factors that affect VR learning. The topic and its representative document emphasized the importance of usability and the design of learning objectives. VR should be easy to use and designed with the user in mind, as users need to understand the usefulness of VR [53]. Cognitive load, which can impede learning with VR immersion, was the focus in *cognitive load and immersive VR* topic. To reduce the effect of cognitive load, it is important to understand the learner: Spatial information and learning strategies are important, objectives and tasks need to be clear, and relevant information should be emphasized [54]. Different stakeholders can also have an impact on VR acceptance as discussed within the topic of *immersive VR in education and learning*. For instance, in school administration and IT, various issues can affect the economic and other support needed for VR. For students, their characteristics, learning outcomes, teacher's role and responsibilities, and learning assessment need to be taken into account to get the most of out VR in education [55].

Most of the topics in this tree were focused on the many different learning spaces and subjects of VR learning. Immersive virtual laboratories and their use arose in the topics *Engineering education in VR*, *Chemistry education in VR*. Complex virtual laboratories can improve lab navigation, lessening the need for assistance and having overall positive impact [56]. iVR laboratories could also involve the students more, motivating them to make learning meaningful and complex concepts easier to understand [57]. In addition, both studies mentioned the need for better results that were more thoroughly validated. Science learning and immersive complex environments were also discussed in *Learning games in VR* in which an immersive role-playing VR game (RPG-VR) was compared with another simpler VR game.

The RPG-VR solution motivated extrinsic learning with self-efficacy, however, with regard to learning achievement, the results were mostly the same; more instructional assistance or other factors may need to be considered for better results [58].

Immersive experiences and environments were also a focus in the cultural and art-related learning in the *VR in cultural learning* topic. The content included an active learning VR game for art history that created an immersive heritage building and cultural experience; it demonstrated that immersion, flow, and engagement in VR could make cultural learning more enjoyable [59]. Architectural design learning was a part of the topic *Study design and learning in VR*, in which it was noted that VR could help students to visualize, design, and review architecture projects; it made the learning of theory easier and immersive [60]. *VR in art and design teaching* had an example of VR use in cross-cultural communication and the development of cinematography education, alleviating the expensive costs of equipment. VR was found to be engaging and especially helpful with the understanding of the filmmaking set, even though it cannot yet replace the unpredictability of a real set [61].

VR was also described in language and sports learning. In *VR in language learning and teaching*, the representative document focused on how students could improve their English learning for one semester in a virtual metaverse environment [62]. The framework for this focused on constructivism and multi-modal teaching, and VR was found to be beneficial as long as the teaching materials were well designed and the experiences were authentic. Even physical training and sport emerged in the *Sports training in VR* topic. VR interested the students, helping their concentration, but it was noted that VR was hindered by its costs, the immaturity of the technology, the possibility of weakening social skills, and a tendency for teachers to rely on the technology too much. Thus, more scientific and comprehensive methods are needed in the future.

Healthcare and training in industries as a broader concept were also described. The largest topic *VR medical simulations education* was formed after removing outliers, and thus, it was a general VR topic represented by words such as "training," "simulation," "medical," "vr," "surgical", and "education". The representative document was on VR's impact on nursing students and it seemed that VR was able to significantly increase nursing students' theoretical knowledge, practical skills, skill retention and satisfaction levels when compared to conventional teaching methods; however, the results regarding critical thinking skills were not significant [63].

For industries, VR was demonstrated in aviation simulations in *VR training in industries*, where the representative document analyzed risky and expensive aviation maintenance and combined this with an interactive immersive training platform for VR [64]. VR can make training more stable and efficient, thus improving the quality of training and person-

nel. In *VR safety training*, the example document combined a collaborative iVR, n-iVR, and blockchain incentives in a web-based framework, whereupon the completion of training scenarios tokens were rewarded to promote performance. This motivated regular training, improving safety awareness and knowledge and assisted in creating better scenarios [65].

As can be seen from these topics, VR can be used in almost any area and in many immersive ways. However, for VR to be more widely accepted, its real benefits need to be more clearly determined and promoted. Usability is naturally important and is the focus of rapid development. As with all learning, learners and their needs are important considerations, especially in immersive interactive environments. In these environments, learning can be gamified; existing examples include reducing pollution in lakes and using 360-videos to teach complex concepts [57]. VR laboratories and VR in general can be used to not replace, but enhance the use of a real environment, as in the case of laboratory navigation [56] or working on a movie set [61]. Again, there is a clear connection to learning by doing and especially on focusing on the learner's needs and environment personalization.

Despite these interesting and innovative solutions, VR demands a great deal from the learner and teacher, and it seems that often the results have not been significant or have pertained to only some aspects and not others. In many cases, other factors or situations hindered the experience. Sometimes, obstacles were related to the technology used or how individuals were not familiar with the technology or the educational methods employed. VR as a learning tool needs more valid and comprehensive data to inform the creation of better standards, materials, and frameworks for learning.

VI. DISCUSSION

This two-phase review focused on recent developments in VR research. A meta-umbrella review was conducted on VR in education in general and how it has been conceptualized. A topic model was created to analyze recent primary VR research topics related to learning. The use of VR varied from simpler learning experiences to building understanding of more complex concepts, such as social learning or healthcare simulations. VR topics in the meta-umbrella review mainly consisted of healthcare with some focus on learning. In the topic model analysis, more varied areas could be seen from specific learning areas, such as chemistry, filmmaking, and worker training. Most of the experimental aspects that were covered in the large set of analyzed literature followed the classical learn-by-doing principle [66].

A. HOW HAS VR BEEN CONCEPTUALIZED IN RECENT UMBRELLA REVIEWS?

The concept and considerations of VR varied in research, and this was especially evident in the umbrella reviews. In general, VR was considered immersive with a possibility for deeper interaction than in traditional education or

learning [31], [41]. VR environments have multisensory elements through different technologies, whether they are HMDs or mobile phones. Customization, feedback, and personalization for the learner are also important, and all these elements affect the level of immersion.

VR definitions can be a bit muddled from a research perspective. This is understandable because at times, anything that happens on a screen, whether it involves video games or virtual worlds with various levels of interaction, can be interpreted as a type of VR; at times, even simple user interfaces were considered to implement VR [32], [35]. This also stems from the varied history and definitions used in VR research. Nowadays, especially in the public eye, most individuals would only think of iVR as real VR. This wide categorization of VR makes it difficult to isolate the results of modern iVR, especially when a broader definition or simpler solutions are included. The wide definition of VR with different implementations, technologies and research areas, result in competing results that could impede each other leading to confused conclusions and a large amount of hard to analyze evidence. This broad definition and multiple combined technologies have been discussed in recent research [67]. Interestingly, the entries in the topic model seemed to indicate a better focus on iVR or at least on more conventional VR, with mentions of immersive role-play situations and more complicated experiments with interaction and engagement.

Pedagogical aspects were a bit difficult to find in the literature because most reviews were more rehabilitation related so the learning processes were examined in a wider sense. Rehabilitation emphasizes the importance of learner and personalization of the process and thus are very learner centered and learning-by-doing oriented [37], [38]. The topic model analysis was more focused on learning and also had a natural learner-centered focus [49], [53], as indicated by previous research [68], [69]. VR seems to especially support this natural learning. Interactive elements and motion supported movements in environments promote learning-by-doing and solutions such as virtual laboratories that were mentioned in the topic model can facilitate this further. However, this demands expertise from multiple stakeholders, including teachers and developers, so design and implementation concerns are important to take note of. As technology advances, VR pedagogy is evolving rapidly, demanding ongoing development and adaptation.

For VR in education, interestingly the focus of the umbrella reviews was more on rehabilitation than on education or training. In contrast, the topic model revealed a broader range of topics and fields. Although previous research has identified similar areas [70], [71], this may suggest that VR research has diversified in recent years. In the future, smaller isolated implementations may give way to standardized VR platforms, particularly in school contexts where subjects could be easier to simulate and even integrate together. Some platforms already exist, such as 3D virtual exhibitions [72] and frameworks for creating immersive experiences [73]. While

highly specialized simulators, such as those of aviation, will likely remain, broader creation tools are becoming more accessible and may support wider adoption across many domains.

In research, the many benefits of VR have been stated multiple times. It can immersively teach, motivate, and provide more control of an environment, which helps learners to grasp difficult concepts with ease. VR is also easy to access and convenient, especially for rehabilitation or more complex simulations, though this impact varies. Common aspects of VR, such as immersion, interaction, feedback, and personalization, were examined in different domains with varying results, which were rarely clearly defined or analyzed [33], [36], [40]. The cases were often limited in their evidence due to low quality, a lack of long-term data, or insufficient sample sizes [33], [35]. This trend appears to continue in the most recent research of the topic model [67], [74]. These aspects can hinder VR research. While the results were often positive, they were frequently inconclusive or lacked evidence for meaningful conclusions on VR's impact on learning. Long-term studies with larger sample sizes are needed to address issues such as the novelty effect or comparisons between different technologies, to help with standardization between methods.

B. WHAT ARE THE MAIN TOPICS IN RECENT PRIMARY STUDIES?

The recent primary studies indicated the rapid pace of VR learning-related research and how it has spread to many different areas. In previous research, VR has been used in all types of areas from rehabilitation [75], [76], [77] to varied learning and training domains [69], [78], [79], [80]. VR has also been used in conjunction with various modern technologies, such as machine learning, motion tracking, emotion and facial recognition, and virtual twins; these combinations further enhanced the VR experiences and made working or learning easier and more motivating. The metaverse and its immersive elements have also been linked to VR [44], examples of which were included in the topic model. This has often been combined with education and modern teaching techniques for experimental learning [62].

For the individual, VR was especially important in a rehabilitation context, and the topic model had examples of immersive personal musical portraits [47] and develop empathy learning [52]. It has been used to advance the social abilities of autistic children [49] and to enhance limb function [50] through VR therapies and rehabilitation, among others. These solutions especially emphasize the importance of interactive and immersive personalized environments in VR learning and how they can be used in the future to further make learning more personal for individuals.

In addition to rehabilitation, VR was also examined across a range of specialized contexts. Acceptance of VR and the user experience emerged in complex environments such as VR engineering [56] and chemistry laboratories [57]. These

environments even help with real world navigation, making students more involved in learning. Gamified immersive solutions, such as role-playing, can enhance a traditional laboratory context [58], making them more engaging. VR can even help with communication breaking cross-cultural boundaries in filmmaking education [61]. In industrial settings, VR is used in advanced aviation simulations [64] and can be combined with technologies such as the blockchain to incentivize safety training participation [65].

Virtual reality research is evolving at a rapid pace, and there is an abundance of evidence for the many qualities of meaningful VR. However, a lack of long-term and thorough research was pointed out in both research phases and has been prominently mentioned in other recently published papers [74]. Interventions are often specific to one-time experiments, which are useful but limited in terms of generalization. Better technology has provided new use cases in various areas as iVR has become more accessible. However, there was a lack of consistent evidence in the many fluctuating areas in which VR can have an influence, and technological acceptance can have a huge impact on the use of VR [9]; this was shown in both phases of the research [39], [53].

The absence of a unified definition for VR can greatly affect VR research. Both phases of the review found issues from simpler solutions to differing definitions of VR, though the topic model suggested some progress. This lack of a clear definition impacts key aspects such as study design and replication. In complex experiments, this variation makes it difficult to define and measure concepts such as immersion complicating the design process. Replication also suffers, making comparisons across studies with differing VR solutions or other instructional methods less reliable. Without unified definitions, VR can be confused with simpler technologies, such as basic virtual interfaces or websites, leading to inconsistent and misleading findings.

In the future, frameworks could help standardize VR research and practices and expand the possibilities of VR through integration of VR education with sound pedagogy. Some studies have also noted this need for more pedagogical focus, thus focusing their framework on constructivist meaningful learning through practice [81]. Experiential and Game-based learning, where the emphasis is on feedback, challenges and rewards, and collaborative learning in shared spaces, also align well with VR learning [82]. Both the meta-umbrella review and topic model also highlighted this natural learning-by-doing principle. However, many frameworks are quite specialized, limiting their broader application [58], [65], [69] focusing more on specific areas such as personalization [83]. Over time a consensus is bound to emerge, but right now, the landscape is fragmented and in development.

Both the meta-umbrella review and the topic model identified several key developments in the use of VR in education. Aspects, such as immersion and sense of presence, were consistently emphasized, though definitions and focus

varied, particularly across umbrella reviews. Learning in VR was seen as a natural experience where the learner is in the center and is taken into account through aspects such as feedback and personalization. VR applications span across diverse subjects from rehabilitation to aviation, highlighting its broad educational potential. However, implementation remains complex for stakeholders such as educators, designers, and learners. Future research should prioritize a clearer definition of VR and base learning on established theories and establish precise goals during the analysis phase to produce more applicable and valid results for future experimentation.

VII. CONCLUSION

The research field of VR in education and learning is vast and versatile. Implications and research activities in many domains are apparent because of the changing and ever-improving possibilities of the level of immersion due to technology developments. VR can be used for learning with abstract or literal concepts; they include rehabilitation and simulations and apply to the classroom and the laboratory. Broader, simpler use cases of VR are still prevalent, but iVR is on the rise. Still, for it to become commonplace, more definitive research is needed.

One of the biggest limitations of this article is the fact that VR research as a subject was widely studied but quite varied. The amount of VR research in one database for one year, even when limiting it to a learning context, was over 6,000 publications. Both VR research and technology move at a fast pace, so this makes it hard to keep up and establish a clear view of the developments, even with a focus on more high-level umbrella reviews. Both the current meta-umbrella review and topic model analysis were also limited by the databases that were searched; there were bound to be some gaps in the retrieved results, even with a thorough search. However, this type of two-phase review is still quite unique, so it offers some new insights in this area.

The wider definition of education, which included rehabilitation and treatments, may also have an effect on the learning perspective of this review, making it more conceptual at times and more difficult to grasp. However, this approach provides new insights into learning and education from a wider, largely unseen perspective. In the topic model, some entries were outliers or more general topics and were not considered significant in the analysis. The analysis on frameworks and lack of standards could also be strengthened by focusing on a deeper analysis of specific educational frameworks such as TPAC. Nevertheless, both the topic model and meta-umbrella review should shed some light on mainstream VR research, its issues, and possible future areas of interest on the whole.

The benefits of VR are multiple in many different rehabilitation and other learning and education situations, but often these results or effects were not fully clear. VR seems to only affect some specific aspects, such as motivation or the treatment of phobias, but not stress, and sometimes even these results differ based on the methods used. The use of

VR usually exhibited some benefits, but they may have been temporary, such as the results of a novelty effect. VR could have a big impact on making learning more personal and important to individuals. Especially for complex concepts, such as social or emotional learning, the potential of VR has only been briefly touched. As VR and related technologies continue to evolve, new opportunities are emerging. One particularly promising facet of this is AI, which could enhance immersion by enabling more dynamic interaction and experimentation. For example, AI characters could provide learners with more realistic dialogue and feedback, supporting deeper personalized learning. The concept of VR itself needs to be clearly defined and compared to other possible methods and technologies. Standards and frameworks need to be formed, and long-term data are crucial to reach the true benefits of VR in education and beyond.

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